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To cite this article: Adithya Vemuri *et al* 2022 *J. Phys.: Conf. Ser.* **2265** 022014

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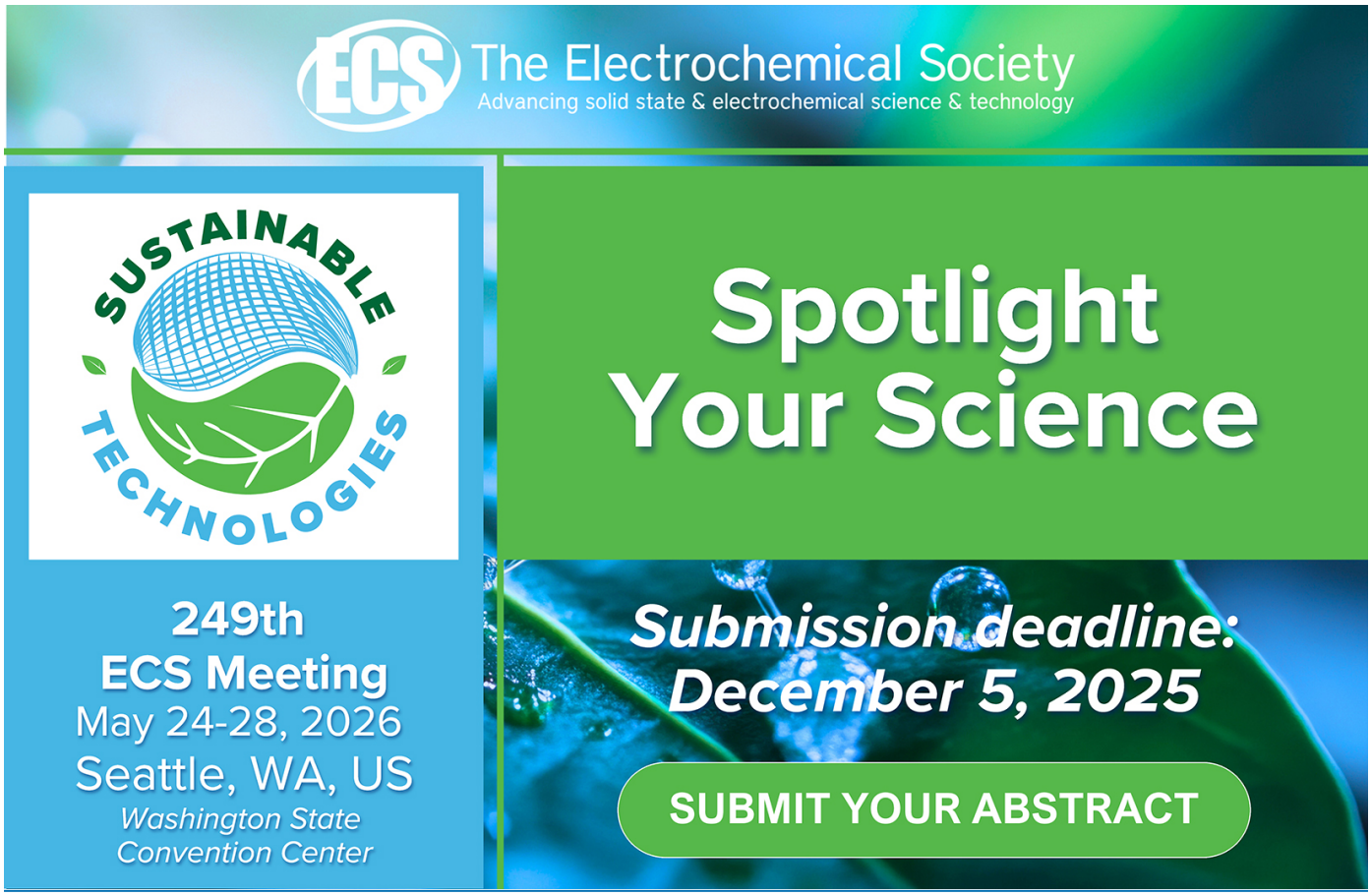

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Modeling extreme weather events for offshore wind in the North Sea: a sensitivity analysis to physics parameterizations in WRF

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Abstract.
Extreme weather events such as storms, cold fronts, and mesoscale convective systems, are capable of producing extreme and sudden precipitation, strong wind gusts and fast changes in wind direction, which are potentially harmful for the operation, power production and maintenance of wind farms. This study aims to provide insight into the modelling of three such events for the offshore wind farms located in the Belgian North Sea. This work extends the ongoing work on generalizing WRF model sensitivities to multiple extreme weather events. This sensitivity analysis considers three cases consisting of Storm Ciara, a low-pressure system and a cold front. These events are modeled using five model setups comprising different combinations of WRF physics parameterizations. The simulated wind direction and speed are evaluated against 10-min averaged SCADA data. The results indicate no single best model setup to accurately simulate all events. The combination of physical parameterization is observed to play a major role in model sensitivities. The use of scale-aware physical parameterization show potential for better performance but is heavily influenced by the combination of model physics and the type of event. No conclusive set of inferences are obtained from a qualitative evaluation of precipitation against observed radar data.

1. Introduction

Accurate prediction of Extreme Weather Events (EWE) plays a key role in wind power forecasting and directly impacts the operational lifetime and maintenance of wind farms. High-impact EWE such as low level jets, high wind shear, high wind veer and wind gusts are experienced by wind farms frequently [1]. This work aims to model high-impact EWE that are found to be potentially harmful for the wind farm operation and maintenance in the Belgian North Sea.

The quite revolution in mesoscale modelling to enable high resolution accurate weather forecasts [2] paves a realistic and tangible approach to understand, analyze, and establish wind energy assets. The use of mesoscale modelling and numerical weather prediction tools are of increasing interest in various fields of wind energy; in wind resource assessment [3], wind farm operation and maintenance [4], simulating extreme weather loading [5] and leading-edge erosion [6]. Literature also establishes the volatility and sensitivity of mesoscale models to simulate

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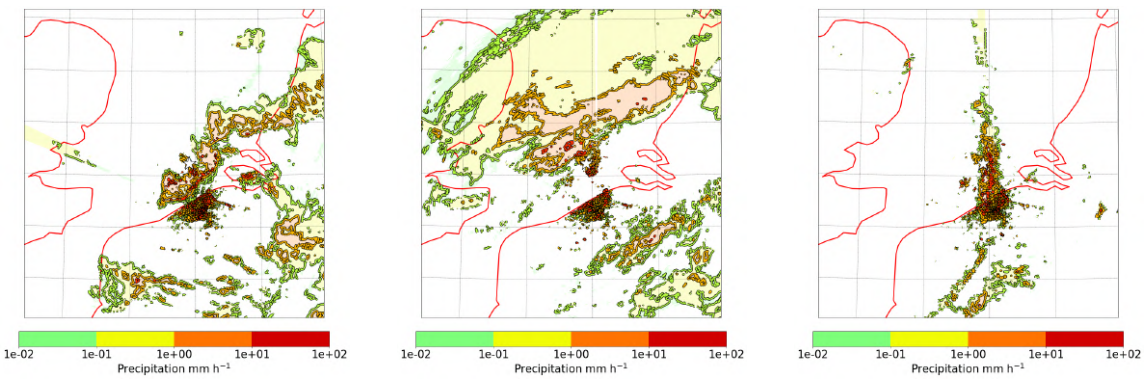
local weather systems, as indicated by Stergiou et al. and Mooney et al. [7, 8] for the European continent. Typically, a mesoscale model setup is established through a sensitivity study of physics and model parameters [9, 10], with a single-event sensitivity study being the common approach [11]. The objective of this work is to provide an insight into the reliability of predicting EWE through a model setup established via a single-event sensitivity analysis, when applied to multiple EWE. More specifically, the generalizability of a recent single-event sensitivity analysis based on Storm Ciara [12] is assessed by applying identified model setups to multiple different EWE observed over the Belgian offshore wind farms. The public domain Weather Research and Forecasting (WRF) [13] mesoscale modelling framework is utilized in this study. The subsequent WRF setups include scale-aware and non-scale-aware physics parameterizations subject to three EWE identified to be potentially harmful for the operation of offshore wind farms located in the Belgian North Sea.

2. Description of the events

The events in this study are selected based on the occurrence of fast changes in wind direction leading to extreme yaw misalignment and subsequent grid-loss as observed by an offshore wind farm located in the Belgian offshore concession zone. The methodology followed to identify these events follows Hannesdóttir et al. [14] to include yaw error. The wavelet analysis identifies anomalies in wind direction change based on a minimum threshold and accompanied by extreme yaw misalignment. Yaw misalignment is found to be a potential cause for the degradation of theoretical lifetime and increased fatigue loading on a wind turbine [15, 16, 17], therefore highlighting the importance of accurately predicting these EWE. The SCADA analysis for the identification of these events includes confidential data that is currently data-protected under a non-disclosure agreement; therefore, no further details can be provided herein. Comparing to IEC 61400-1:2005 definition [18] of extreme coherent gust with wind direction change (ECD), the Storm Ciara, low pressure system and cold front events lie within the range of defined fast transients, i.e. with a wind-direction change of 33°, 48° and 65° respectively over a period of 10 s.

The case studies include a microburst occurring during Storm Ciara on 10 February 2020, a low pressure system occurring over the Belgian North Sea on 24 December 2020 and a cold front on 27 June 2020. Figure 1 presents observed RADAR reflectivity plots for the aforementioned events at relevant timestamps of observed fast changes in wind direction.



(a) (b) (c)

Figure 1: Extreme weather events considered as case studies (a) storm Ciara, (b) low pressure system, and (c) cold front.

2.1. Microburst during Storm Ciara

An extratropical cyclone, Storm Ciara, was experienced by the Belgian coast during the second week of February, 2020. During the storm passage, a bow echo (presented in fig. 1a) was observed at 04:40 UTC on 10 February 2020, indicating the presence of a microburst phenomenon [19]. The fast changes in wind direction and yaw error are accompanied by a strong precipitation peak as observed in radar data and verified by the wind profiler located within the wind farm, presented in fig.2.

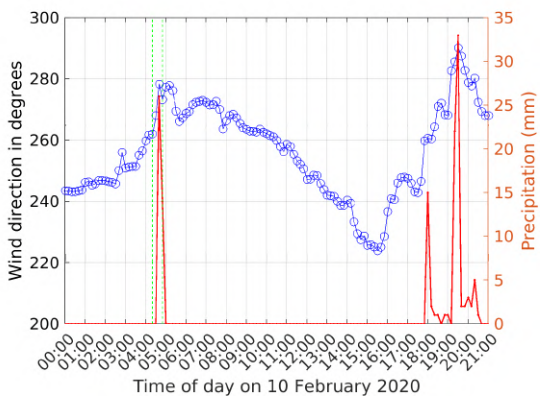


Figure 2: Precipitation observed by the offshore wind farm plotted against 10-min averaged wind direction, highlighting the period of interest (highlighted in green) at 04:40 UTC accompanied by sudden precipitation.

2.2. Low pressure system

On 24 December 2020, fast changes in wind direction followed by undesirable turbine behaviour were recorded by an offshore wind farm at 02:00 UTC. Synoptic maps [20] from this time period indicate the presence of a fully developed low pressure system passing over the Belgian offshore concession zone. Moreover, RADAR reflectivity plots at the time indicate the presence of large precipitation cells (fig. 1b) bordering the zone. Similar to the Storm Ciara event, precipitation was recorded by the offshore wind farm accompanied by the fast changes in wind direction (not shown for brevity).

2.3. Cold front

On 27 June 2020, a cold front was observed [20] over the Belgian offshore wind farms accompanied by sudden precipitation as observed by the offshore wind farm. The observed RADAR reflectively is presented in fig. 1c, precipitation cells were observed during the afternoon hour at 15:30 UTC. A decaying low pressure system was observed over the British isles with its leading edge bordering the Belgian offshore concession zone.

3. Methodology and model setup

The 3 EWE introduced in section 2 are simulated using a set of five WRF setups including 2 of the best-case and 2 of the worst-case setups from Vemuri et al. [12]. Furthermore, an additional setup using the three-dimensional PBL parameterization defined by Zhang et al. [21] is included.

This study utilizes the WRF model version 4.2.2 to maintain consistency with Vemuri et al. [12]. All simulations were initialized with hourly reanalysis dataset from ERA5 [22] with a 24-hour spin-up period and 20 s time-step. The common model parameters along with the combination of simulation runs are summarized in table 1. The model domain configuration is presented in fig. 3. No wind farm parameterizations are considered in this study. 2 of the

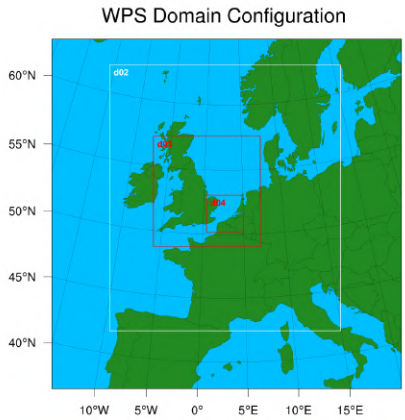


Figure 3: WRF domain configuration considered.

Table 1: WRF model setup, parameters varied in this sensitivity study are shown in italics. Scale-aware physics parameterizations are underlined.

Numerical setup	
Nested domains	4
Horizontal resolution	27 km (d01) × 9 km (d02) × 3 km (d03) × 1 km (d04)
Terrain following vertical levels	57
Model top pressure	1000 Pa
Time-step	20s
Spin-up and simulation period	24h + 21h
Initial & boundary conditions	1h ERA5 reanalysis
Physics parameterizations	
Radiation	RRTMG
Land surface	unified Noah land-surface
Simulation run 1	*MYNN / Thompson / KF*
Simulation run 2	*Shin-Hong / WSM5 / KF*
Simulation run 3	*Shin-Hong / Morrison / msKF*
Simulation run 4	*Shin-Hong / Morrison / GF*
Simulation run 5	*Zhang / Morrison / GF*

best-case model setups identified in Vemuri et al. [12] both using a scale-aware Shin-Hong 1D PBL parameterization [23], combined with the Morrison microphysical scheme [24]. They are distinguished by the cumulus scheme, using a multi-scale Kain-Fritsch (msKF) [25] and Grell-Freitas (GF) parameterization [26], respectively. Similarly, two of the worst-case model setups both use the Kain-Fritsh (KF) [27] cumulus parameterization, and are distinguished by the use of a non-scale-aware MYNN 1D PBL [28] with Thompson [29] microphysics, and Shin-Hong PBL coupled with WSM5 [30] microphysics, respectively. The model setup with 3D PBL was run with the WRF setup utilizing GF cumulus and Morrison microphysics parameterization. KF and GF cumulus parameterizations were explicitly turned off in domain 3-4 and domain 4 (corresponding to scheme definitions), respectively.

The WRF simulations are compared to 10-min average wind direction and horizontal wind speed from SCADA data from an offshore wind farm located in the Belgian North Sea, and to qualitatively to reflectivity data provided by a dual polarization C-band RADAR from the Royal Meteorological Institute - Belgium (RMI-B) located at Jabbeke on the Belgian coast. The metrics Mean Absolute Error (MAE) and Kantorovich distance are utilized for the evaluation of wind direction and wind speed. The definition of Kantorovich distance as described in Wang et al. [31] is utilized in this study. To establish a single metric, the MAE and Kantorovich distance are normalized over each of the five model setups, and combined into a so-called

Normalized Euclidean Distance defined as, NED $= \sqrt{\text{MAE}^2_{norm.} + \text{Kantorovich}^2_{norm.}}$. Where $\text{MAE}_{norm.}$ and $\text{Kantorovich}_{norm.}$ represent the normalized metric-specific values for all event-specific simulations. Further, average NED is defined as the mean of wind direction and speed NED. The metrics are evaluated for a total of 21 hours for each event. The simulated RADAR reflectivity is qualitatively compared to observed reflectivity data from RMI-B for the timestamps described in the section 2 to ascertain a possible tuning of WRF to produce better wind results at the expense of precipitation and vice-versa. Reflectivities are compared to rain fall using the Marshall-Palmer relation [32].

4. Results and discussion

The following sections present the results for individual WRF simulations. Following performance evaluation metric tables are sorted by the increasing order in fidelity of physics schemes identified for Storm Ciara in the preceding study by Vemuri et al. [12], with runs 1-2 as 2 of the worst-case WRF setups, runs 3-4 as 2 of the best-case WRF setups (see section 3). For brevity purposes, precipitation plots are presented for runs 1, 4 and 5. The results for individual case studies are discussed in designated sub-sections.

4.1. Storm Ciara

The performance evaluation metrics for the case of Storm Ciara are presented in the table 2. The dataset is largely akin to Vemuri et al. [12]. The overall best performance was observed by run 5. Runs 3-4 outperformed runs 1-2 by 20% (considering total average NED of both runs), the former utilizing scale-aware PBL and cumulus parameterizations with higher fidelity microphysics. The latter utilize combinations of scale-aware and non-scale-aware PBL schemes coupled with lower fidelity in microphysics and non-scale-aware cumulus parameterization. Qualitatively, ensemble results show a good match with SCADA for both wind direction and wind speed time series as shown in fig. 4. A comparison of the individual run-by-run time-series (not shown for brevity) indicates runs 3, 4 and 5 to better capture the overall trend in SCADA wind direction and wind speed in comparison to runs 1-2.

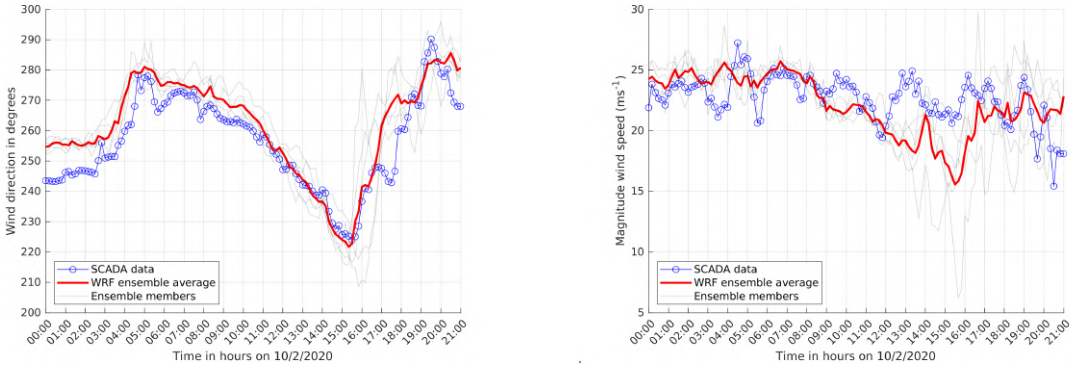
Table 2: Performance metric table for the case study of Storm Ciara on 10 February 2020. Evaluation was performed against 10-min averaged SCADA data using MAE and Kantorovich distance.

					Wind direction			**Wind speed**			
#	*ERA5 res.*	*PBL scheme*	*Microphysics scheme*	*Cumulus scheme*	MAE (deg)	Kantorovich (cost)	Norm. euclidean distance	MAE (deg)	Kantorovich (cost)	Norm. euclidean distance	Average NED
1	1h	MYNN	Thompson	KF	9.26	5.83E+03	1.56	2.72	6.41E+03	1.65	**1.61**
2	1h	Shin-Hong	WSM5	KF	7.68	1.93E+04	3.45	2.47	6.75E+03	1.62	**2.53**
3	1h	Shin-Hong	Morrison	msKF	7.17	8.84E+03	1.78	1.89	5.10E+03	1.23	**1.50**
4	1h	Shin-Hong	Morrison	GF	7.17	9.14E+03	1.82	2.67	7.46E+03	1.77	**1.79**
5	1h	Zhang	Morrison	GF	8.69	9.45E+03	1.97	1.84	3.10E+03	0.98	**1.48**
6	Ensemble				6.19	9.06E+03	1.75	2.00	5.13E+03	1.27	**1.51**

A qualitative comparison of observed and simulated RADAR reflectivity is presented in fig. 5. The simulated reflectivity is observed to vary strongly from setup to setup, qualitatively marginal improvements were observed for simulation run 5. Run 5 also performing the best for wind variables. However, this qualitative comparison allows for few statements to be drawn on precipitation modelling fidelity in this study.

4.2. Low pressure system

The performance metrics obtained for the case of low pressure system are summarized in table 3. The best case WRF setup is identified to be run 2, considering average NED as the evaluation



(a) (b)

Figure 4: Time series comparison between WRF ensemble average and SCADA data during Storm Ciara on 10 February 2020, on (a) wind direction and (b) wind speed. Ensemble members are presented in grey.

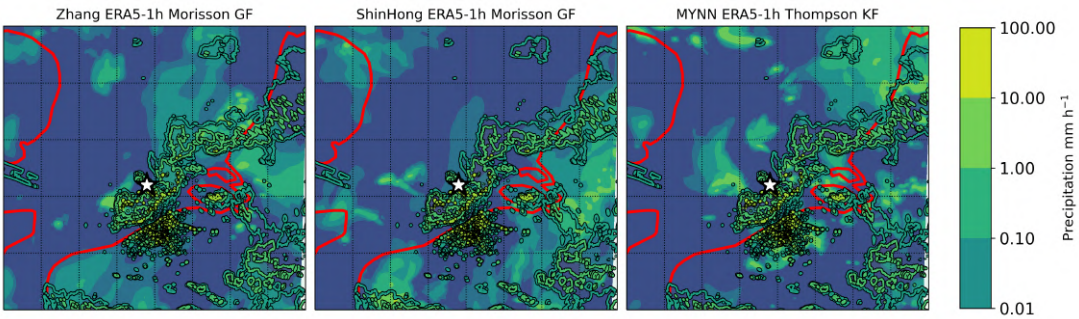


Figure 5: Simulated and observed RADAR rainfall rate for a bow echo observed over Belgian North Sea on 10 February 2020 at 04:40 UTC, WRF simulations are presented in contour fill and RADAR plots in line contours. Belgian offshore wind farms are indicated by a star symbol.

metric. Runs 1, 3 and 5 result perform worse in comparison to runs 2 and 3. The best-case run 2, comprises scale-aware Shin-Hong PBL, coupled to non-scale-aware KF cumulus parameterizations and WSM5 microphysics. Overall, the accuracy of results for simulation runs 2, 3, 4 and 5 is observed to be similar both, qualitatively and quantitatively. The ensemble average plots presented in fig. 6 indicate an overall similar wind direction and wind speed predictions for all considered simulations, albeit with a time lag for wind direction and an under prediction of wind speed during the early hours.

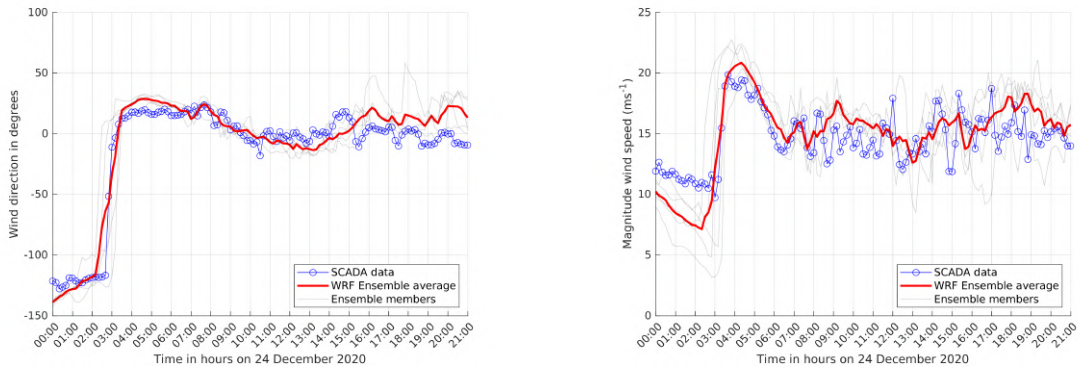
The qualitative comparison of simulated RADAR reflectivity and observed is presented in fig. 7. The case of the low pressure system observed heavy rain fall accompanying large precipitation cells. The WRF simulated RADAR reflectivity for all cases is found to be qualitatively similar to the observed RADAR data.

4.3. Cold front

The results for the performance metrics for the case of cold front are presented in table 4. Considering average NED as the evaluation metric, run 5 produced the best overall results. With runs 2 and 3 resulting in the worst NED values. The ensemble average performing poorly, similar to run 2. The time-series plots for wind direction indicate qualitatively poor performance by runs 1, 2 and 3, to effectively capture the transience in wind direction and an overestimation

Table 3: Performance metric table for the case study of low pressure system on 24 December 2020. Evaluation was performed against 10-min averaged SCADA data using MAE and Kantorovich distance.

					Wind direction			**Wind speed**			
#	*ERA5 res.*	*PBL scheme*	*Microphysics scheme*	*Cumulus scheme*	MAE (deg)	Kantorovich (cost)	Norm. euclidean distance	MAE (deg)	Kantorovich (cost)	Norm. euclidean distance	Average NED
1	1h	MYNN	Thompson	KF	16.09	5.68E+04	2.36	3.20	5.68E+03	2.03	**2.19**
2	1h	Shin-Hong	WSM5	KF	10.67	1.61E+04	1.03	1.90	1.63E+03	0.95	**0.99**
3	1h	Shin-Hong	Morrison	msKF	11.86	2.55E+04	1.31	2.38	2.53E+03	1.25	**1.28**
4	1h	Shin-Hong	Morrison	GF	11.88	1.73E+04	1.14	2.11	6.19E+03	1.82	**1.48**
5	1h	Zhang	Morrison	GF	12.32	3.97E+04	1.70	2.03	4.35E+03	1.42	**1.56**
6	Ensemble				10.62	1.89E+04	1.08	1.68	3.57E+03	1.17	**1.13**



(a) (b)

Figure 6: Time series comparison between WRF ensemble average and SCADA data during low pressure system event on 24 December 2020, on (a) wind direction and (b) wind speed. Ensemble members are presented in grey.

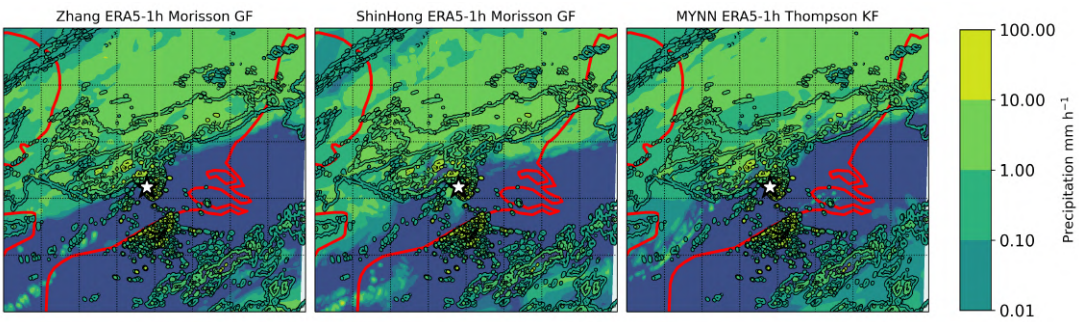
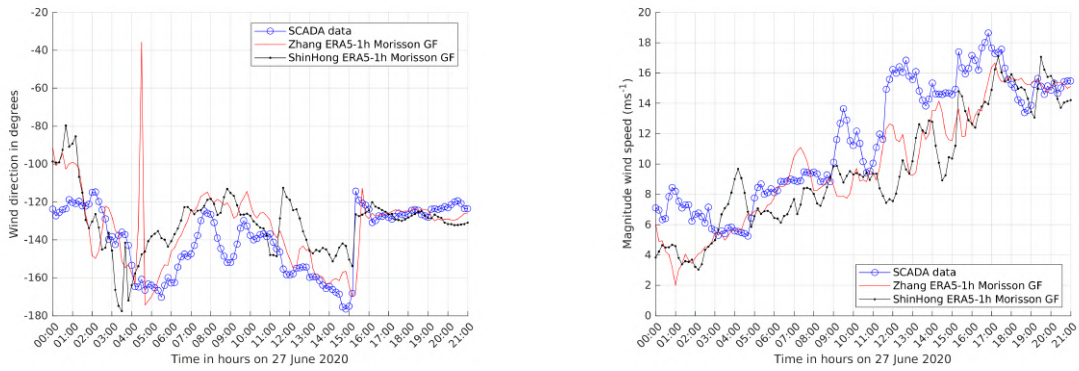


Figure 7: Simulated and observed RADAR rainfall rate for a low pressure system observed over Belgian North Sea on 24 December 2020 at 02:00 UTC, WRF simulations are presented in contour fill, RADAR plots in line contours. Belgian offshore wind farms are indicated by a star symbol.

during the early hours, followed by an underestimation during the period of interest at 15:00 UTC. The qualitative evaluation of time-series plots for run 5 against run 4 are presented in fig. 8. The fast changes in wind direction at the period of interest are better captured by run 5. However, the wind direction changes observed during the early hours is captured relatively poorly by all simulations (as indicated in the ensemble plots, fig. 10). Wind speed results, indicate a slow day-long ramp relatively under-predicted by all WRF simulations.

Table 4: {Performance metric table for the case study of cold front on the 27 June 2020. Evaluation was performed against 10-min averaged SCADA data using MAE and Kantorovich distance.

					Wind direction			Wind speed			
#	*ERA5 res.*	*PBL scheme*	*Microphysics scheme*	*Cumulus scheme*	MAE (deg)	Kantorovich (cost)	Norm. euclidean distance	MAE (deg)	Kantorovich (cost)	Norm. euclidean distance	Average NED
1	1h	MYNN	Thompson	KF	20.17	6.36E+04	1.19	1.98	2.40E+03	0.95	**1.07**
2	1h	Shin-Hong	WSM5	KF	25.87	7.85E+04	1.50	2.89	6.30E+03	1.77	**1.64**
3	1h	Shin-Hong	Morrison	msKF	46.79	1.94E+05	3.22	3.04	8.56E+03	2.18	**2.70**
4	1h	Shin-Hong	Morrison	GF	15.47	2.14E+04	0.71	2.50	3.24E+03	1.23	**0.97**
5	1h	Zhang	Morrison	GF	12.17	2.17E+04	0.59	1.95	3.69E+03	1.11	**0.85**
6			Ensemble		21.59	7.81E+04	1.37	2.26	4.58E+03	1.33	**1.35**



(a) (b)

Figure 8: Time series comparison between WRF and SCADA data during cold front event on 27 June 2020 for runs 5 and 4, on (a) wind direction and (b) wind speed.

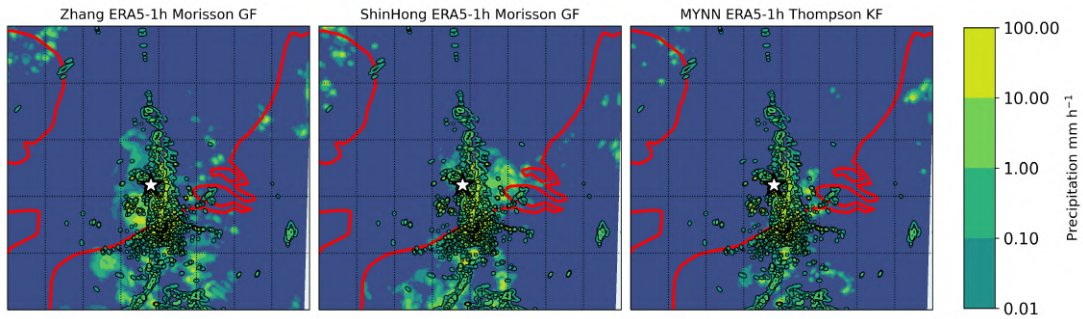
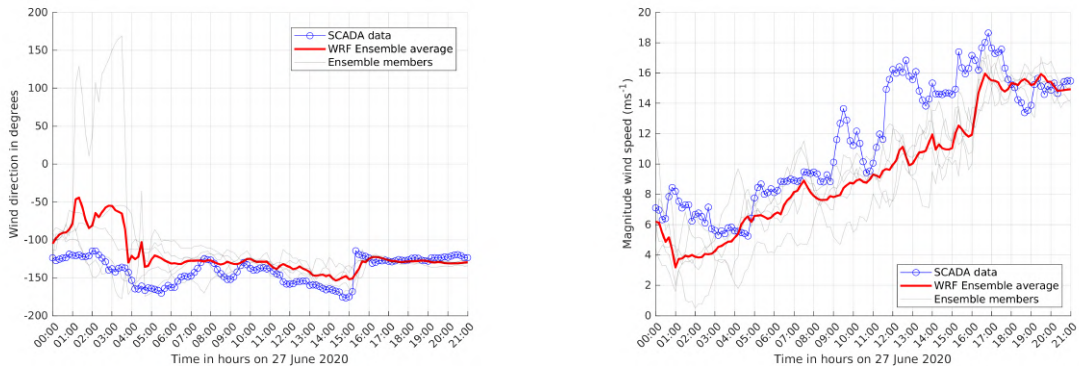


Figure 9: Simulated and observed RADAR rainfall rate of a cold front observed over Belgian North Sea on 27 June 2020 at 15:30 UTC, WRF simulations are presented in contour fill, RADAR plots in line contours. Belgian offshore wind farms are indicated by a star symbol.

The time-series plots for ensemble average are presented in fig. 10 indicating the wind speed transience to be predicted poorly by the ensemble average. Similarly, fast changes in wind direction are predicted poorly by all simulations during the early hours. With run 5 capturing the period of interest better in comparison. Note that the ensemble averaging acts to damp out strong spurious fluctuations in individual members, as observed in the evaluation metric, NED. The results for the observed versus simulated RADAR reflectivity are presented in fig. 9. Qualitatively, precipitation cells were produced by all WRF simulation during the period of interest, with run 5 predicting slightly larger precipitation cells, run 5 is also observed to

produce best case wind variables results this event.



(a) (b)

Figure 10: Time series comparison between WRF ensemble average and SCADA data during cold front event on 27 June 2020, on (a) wind direction and (b) wind speed. Ensemble members are presented in grey.

5. Conclusions and recommendations

The current study outlines the work that has been performed to generalize model sensitivities of WRF simulations over the Belgian North Sea applied to multiple EWE. This study uses 5 WRF model setups to include scale-aware and non-scale aware physical parameterizations. 3 EWE are considered in this study; Storm Ciara occurring on 10 February 2020, a low pressure system on 24 December 2020 and a cold front on 27 June 2020. Model sensitivities are evaluated against 10-min averaged SCADA data using the performance metrics MAE and Kantorovich distance, later normalized as NED. The generalizing of a single-event tuned WRF physics setup, when applied to multiple events is found to produce varying performance of wind direction and speed predictions. Accurately capturing the fast transients in wind direction posed a challenge for WRF simulations, especially in the case of cold front. The combination of WRF physical parameterizations is observed to play a major role in the simulation of EWE. The use of scale-aware physical parameterizations exhibit a potential improvement in EWE simulation however, yields varying performance based on the event, for instance, the case of Storm Ciara and cold front indicate a clear advantage in utilizing 3D scale-aware Zhang PBL scheme combined with scale-aware cumulus and higher order microphysics to better capture wind transience. However, the case of low pressure system scale-aware Zhang 3D PBL scheme under performs in comparison to the scale-aware Shin-Hong 1D PBL coupled with WSM5 microphysics and non-scale-aware KF cumulus parameterization. The identification of a single best model setup for simulating different EWE is likely infeasible given the current state-of-the-art physical parameterizations. The ensemble average estimates provide similar results to the best case setups for EWE Storm Ciara and low pressure system. However, for the case of cold front the best case setup out performs the ensemble average. Indicating the importance of the diversity in simulation setups when considering ensemble average of different EWE.

The qualitative evaluation of precipitation observes accurate prediction of wind direction and wind speed to be correlated with correct modeling of the meteorological event itself. For instance, the case of Storm Ciara and cold front, observe improvements in precipitation results for run 5. Which also predicts the best-case results for wind variables. The dependency between wind behavior and precipitation is again highlighted by the detection of a strong precipitation peak measured by the wind profiler located within the farm, that coincides with the fast change in wind direction presented in fig. 2, this highlighting the importance of properly predicting

precipitation. However, this qualitative evaluation allows for few conclusions to be drawn from WRF simulations evaluated against RADAR data. With simulation run 5, comprising 3D PBL and scale-aware cumulus parameterizations coupled with higher order microphysics parameterization, to be the qualitatively better performer.

In future work, we will build upon the results of this paper to define the optimum combination of physics schemes to be used for the accurate prediction of extreme weather events. The next step will be to extract the dynamic loading conditions exerted on the machines by increasing resolution down to turbine scales. For this, an appropriate modeling of turbulence will require the use of a 3D PBL scheme, for which an in-house developed scale-adaptive scheme [33] will be implemented into WRF. This will provide the realistic aerodynamic loading conditions to be used as inflow to aeroelastic computations, in view of refining the design load cases to be more representative of the weather conditions as experienced in the field.

Acknowledgments

This work has received funding from the Flemish Government through the Agency for Innovation and Entrepreneurship (Vlaams Agentschap Innoveren en Ondernemen, VLAIO) through the cSBO project SeaFD in the context of the MaDurOS program for Material Durability for Off-Shore of the cluster on Strategic Initiative for Materials in Flanders (SIM). Adithya Vemuri would like to thank Pieter-Jan Daems from Vrije Universiteit Brussel for sharing details on the identified extreme weather events.

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